# Response to Reviewers — TCS-D-26-00494

Manuscript: "Distributional Learning of Context-Free Languages under Fixed Finite-Monoid Typing"

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We thank the Editor and both reviewers for their detailed and constructive reports. Their comments led us to correct the learning algorithm, remove an unused typing component, repair the terminal-rule completeness argument, and substantially reorganize the manuscript.

Throughout this response, unqualified section, theorem, proposition, lemma, and corollary numbers refer to the revised manuscript; references to the submitted version are marked "submitted" or "former". Subheadings that reproduce a reviewer's original section grouping (for example, "Section 7" or "Sections 8–10") refer to the submitted manuscript. For ease of reference, we number the Editor's comment as E.1 and the reviewers' comments as R1.1–R1.37 and R2.1–R2.11. Reviewer quotations retain the original section grouping, with only typographical normalization and omissions indicated by ellipses; related remarks are grouped by topic within each section. Each response identifies the revised section or theorem-like statement directly. The locator table below gives the revised statement or section and its verified marked-PDF page and margin-line anchor.

## Overview of changes

**Historical and related-work clarification.** The revised Introduction now cites the author's historical 2014 preprint (arXiv:1409.6247v3), which preceded Yoshinaka (2015) and already proposed substitutability relative to recognizable equivalence relations and a positive-data learner. We also acknowledge Yoshinaka's subsequent framework of signatures with disjoint sorts: the nonempty fibers $h^{-1}(m)\cap\Sigma^+$ of the finite-monoid homomorphism $h : \Sigma^* \to M$ fixed in this paper can be represented as disjoint sorts, with typed concatenation and promotion to a whole-string sort. Thus this is our application of Yoshinaka's general framework, not a choice of $h$ attributed to Yoshinaka; the existence of a positive-data learner has a corresponding qualitative treatment in Yoshinaka (2015, Theorem 1). The current manuscript does not claim independent novelty merely for the existence of a learner or for the classical fixed-window algebraic criterion or the window-signature representation already present in Yoshinaka (2015, Example 4). Proposition 3.3 makes the corresponding finite-monoid morphism explicit. The manuscript's finite-monoid-specific contributions include quantitative encoded characteristic-data bounds (notably the window–finite-group short-yield argument), retyping closure, the linear regime, and explicit separation results. The original 2014 proofs and quantitative claims are not asserted to have been fully correct. Our reference is explicitly to v3, not to the later arXiv v4 record containing a different preliminary revision. During revision, Section 7 was extended from fixed windows to windows combined with finite-group values; this is distinct from the classical local-triviality criterion. Additional related citations include Clark–Yoshinaka (2016), Coste–Garet–Nicolas (2012), and Coste–Nicolas (2019). We also cite Costa (2002, *Semigroup Forum* and *Semigroups, Algorithms, Automata and Languages*) for the previously studied pseudovariety joins involving locally trivial semigroups. Our revised Section 7 uses this classical setting to locate the new quantitative grammar-learning bound. This change is reflected in responses R1.3 and R2.1 below.

The revision makes four central corrections. First, it separates the set-driven reconstruction operator $\mathcal{B}_h$ from the conservative sequential learner $\mathcal{A}_h$, so that the Gold learner now stabilizes syntactically. Second, the unused outer-context type annotations are removed, and the occurrence-indexed hypothesis symbols are quotiented to one substring-indexed symbol $[x]$ per observed factor. The resulting batch construction follows Yoshinaka's substring grammar (as also presented by Clark), replacing the window comparison in the unary rules by equality of $h$-types. Third, the terminal-rule completeness gap is repaired by adding witnesses for typed terminal rules and using the unary Rule (U) before the lexical Rule (L) when necessary. Fourth, fixed-$h$ substitutability is stated for nonempty internal fragments, with $\lambda$ handled separately at the start symbol.

Several claims have been narrowed accordingly. Presentation-order independence is claimed only for the batch operator, not for the sequential learner. Polynomial characteristic-data bounds are stated for the batch operator in the specified quantitative regimes, while the sequential learner is proved separately to have Gold convergence and polynomial-time updates. We withdraw the submitted Section 7 title, *The Linear Subclass: A Full Polynomial Time-and-Data Theorem*, as a characterization of the sequential learner: revised Theorem 8.3 gives polynomial characteristic data for $\mathcal{B}_h$ relative to a chosen linear CFG, but no incremental polynomial-data guarantee for $\mathcal{A}_h$ is claimed. Section 2.1 now explains the difference between the set-based criterion (Eyraud–Heinz–Yoshinaka, 2016, Definition 2.8) and the stronger incremental criterion (ibid., Definition 2.5). The former claim that outer-context types are necessary has been withdrawn, and the separate context-transport rule is no longer needed in the quotient construction.

The revision also strengthens the previous typed-thickness example. Proposition 6.3 now supplies a family of finite Clark–Eyraud substitutable languages $L_n$ represented by grammars of size and ordinary thickness $O(n)$. Under trivial typing $\mathcal{B}_{h_0}$ has polynomial characteristic data, but a fixed two-element semilattice typing forces

every characteristic sample for $\mathcal{B}_{h_{\mathrm{U}}}$ to contain a word of length $3 \cdot 2^n - 2$. Moreover, Proposition 6.3(iv) establishes the same lower bound for any set-driven operator admitting characteristic samples for both $L_n$ and $L_n \setminus \{w_n^*\}$ in the fixed-typing class. Corollary 6.4 gives the sharper, presentation-independent bound $\tau_{h_{\mathrm{U}}}^{\mathrm{typ}}(G) \geq 3 \cdot 2^n - 2$ for every reduced SSBNF grammar of $L_n$, regardless of size. Lemma 7.6 shows $U_1 \notin \mathbf{LI} \vee \mathbf{G}$, so the algebraic condition of Corollary 7.5 cannot simply be dropped. Neither necessity for every typing nor any lower bound on the running time or convergence stage of arbitrary Gold learners is claimed. These results are new in this revision.

The manuscript has also been substantially shortened and reorganized. Basic definitions are consolidated in Section 2, the reconstruction operator is now defined before the target-side refinement used in the completeness proof, the former three boundary sections are merged into Section 9, and technical normalization proofs are in Appendices A–B; the new bracketed-family substitutability argument is in Appendix C. The submitted Appendix A already established intersection and inverse-homomorphism closure facts; these are condensed in Proposition 3.2, whose inverse-homomorphism argument is adapted to the revised nonempty-fragment convention by distinguishing erased factors. Sections 6–8 reorganize the quantitative discussion and answer Reviewer 2's question on the linear theorem through a nonregular example and a general regular-filter construction within the linear subclass. Section 9 is shorter overall and also contains the new congruential-class Proposition 9.7, which was suggested by Reviewer 1. The revised linear bound in Theorem 8.3 may be made explicit, for the normalized linear grammar $G_0$, as an encoded-witness estimate $\|\mathsf{W}(\widetilde{G}_0)\| = O(|G_0|^3|M|^4)$; we do not claim the same degree relative to an unnormalized input grammar. The revisions also clarify the nonerasing inverse-image observation, the factorization $S^n = SE(S)S$, and Lemma 7.1's width count. The revised nonlinear example in Proposition 9.1 defines admissible entry-height sets $\mathcal{Q}(x)$ by canonical completion, states a concatenation identity justified in a footnote, and proves fixed-$h_\star$ substitutability by separating $ba$-containing factors from factors in $a^*b^*$. The sharp bound $\rho \geq 2$ for the capped-counter separation is now accompanied by a footnote explaining $(1,1)$-substitutability of $\mathrm{CTR}_1$. We distinguish newly added material from previously submitted results and from prior results of Eilenberg (1974), Kanazawa (1996, 1998), Yoshinaka (2008), and Pin (1997, 2025). In particular, revised Proposition 3.5 records a known locally trivial-semigroup characterization in positive-image/kernel notation, not a new theorem of finite-semigroup theory. The Introduction also cites Kanazawa (1998, Chapters 5 and 8) for characteristic samples of rigid grammars and finite-elasticity arguments for string languages of $k$-valued categorial grammars. These are complementary positive-data learning results, not results from which our fixed-$h$ theorem follows. No proof or mathematical claim is changed by this bibliographic clarification. The revised preliminaries and Proposition 3.1 now cite Eilenberg (1974, Chapter III, Propositions 10.1–10.2) for the classical syntactic-congruence and finite-monoid characterization of regular languages. This historical attribution changes no mathematical claim; the locally trivial-semigroup statement remains attributed to Pin.

**Selected renumbering guide.** Because the manuscript was substantially reorganized, the following table collects the submitted items most frequently referred to in the reports.

| Submitted manuscript | Revised manuscript |
|---|---|
| Definition 2.3 ($\sim_h$) | Definition 2.1 |
| Definition 3.2 (Clark–Eyraud) | unnumbered special case, Section 3.1 |
| Proposition 3.5 | Proposition 3.3 |
| Theorem 3.7 | Theorem 3.4 |
| Example 4.3 | removed |
| Submitted Rules (2), (3) | quotient construction; unary Rule (U) |
| Lemma 5.2(i) | witness set (2), Theorem 5.3 |
| Theorem 5.7 | Theorem 5.4 |
| Corollary 5.8 | Corollary 5.5 |
| Proposition 7.2 | Proposition 8.1 |
| Theorem 7.9 | Theorem 8.3 |
| Theorem 8.6 | Theorem 9.2 (simplified proof) |
| Corollary 8.7 | removed |
| Propositions 10.3–10.5 | consolidated in Proposition 9.1 |
| Lemma 9.11 | Lemma 9.6 |
| Appendix A (closure facts) | Proposition 3.2 (adapted) |
| $\mathrm{CCL}_p$, CCL | $\mathrm{CTR}_\rho$, CTR |
| Counter alphabet $\{d, u, ;\}$ | $\{\uparrow, \downarrow, \#\}$ |
| $L_Y$ (classical language) | $L_{\mathrm{Luk}}$ (Łukasiewicz) |
| Props. 9.2, 9.6, 9.9; Lemma 9.8 | brief statements/citations in Section 9.3 |

**Locator guide for the additions-marked revised manuscript.** Each location below is anchored to a named section or statement, with its actual PDF page and a margin-line number checked in the freshly compiled

additions-marked manuscript. A section anchor may cover multiple revised paragraphs; the named statement identifies the relevant paragraph.

| Item | Revised-manuscript locator |
|---|---|
| E.1 | Definition 2.1 (marked PDF p. 4, margin l. 170); Sections 4–5 (marked PDF p. 8, margin l. 313) |
| R1.1 | Section 2.1 (marked PDF p. 4, margin l. 137); Section 4.2 (marked PDF p. 9, margin l. 348); Section 5.2 (marked PDF p. 11, margin l. 423) |
| R1.2 | Introduction (marked PDF p. 1, margin l. 23); Appendices A–C (marked PDF p. 22, margin l. 870) |
| R1.3 | Introduction and related work (marked PDF p. 1, margin l. 23) |
| R1.4 | Introduction (marked PDF p. 1, margin l. 23); Definition 2.1 (marked PDF p. 4, margin l. 170) |
| R1.5 | Section 1.2 (marked PDF p. 2, margin l. 81) |
| R1.6 | Sections 9.3–9.4 (marked PDF p. 20, margin l. 795) |
| R1.7 | Section 4.2 (marked PDF p. 9, margin l. 348) |
| R1.8 | Section 2 (marked PDF p. 3, margin l. 112); Section 4.1 (marked PDF p. 8, margin l. 314) |
| R1.9 | Section 5.1 (marked PDF p. 9, margin l. 365) |
| R1.10 | Introduction (marked PDF p. 1, margin l. 23); Section 3 (marked PDF p. 5, margin l. 177) |
| R1.11 | Definition 2.1 (marked PDF p. 4, margin l. 170); Section 3.1 (marked PDF p. 6, margin l. 233) |
| R1.12 | Section 3.1 (marked PDF p. 6, margin l. 233); Proposition 3.3 (marked PDF p. 6, margin l. 244); Proposition 3.5 (marked PDF p. 7, margin l. 292) |
| R1.13 | Section 2 (marked PDF p. 3, margin l. 112); Theorem 3.4 (marked PDF p. 7, margin l. 268) |
| R1.14 | Theorem 3.4 (marked PDF p. 7, margin l. 268) |
| R1.15 | Section 2 (marked PDF p. 3, margin l. 112); Section 5.1 (marked PDF p. 9, margin l. 365) |
| R1.16 | Section 2 (marked PDF p. 3, margin l. 112) |
| R1.17 | Section 5.1 (marked PDF p. 9, margin l. 365) |
| R1.18 | Section 4 (marked PDF p. 8, margin l. 313); Sections 5.1–5.2 (marked PDF p. 9, margin l. 365) |
| R1.19 | Proposition 5.2 (marked PDF p. 10, margin l. 398) |
| R1.20 | Section 5.1 (marked PDF p. 9, margin l. 365) |
| R1.21 | Section 5.1 (marked PDF p. 9, margin l. 365) |
| R1.22 | Proposition 5.2 (marked PDF p. 10, margin l. 398) |
| R1.23 | Proposition 5.2 (marked PDF p. 10, margin l. 398) |
| R1.24 | Section 4.2 (marked PDF p. 9, margin l. 348); Section 5.2 (marked PDF p. 11, margin l. 423) |
| R1.25 | Section 4.2 (marked PDF p. 9, margin l. 348); Section 5.2 (marked PDF p. 11, margin l. 423) |
| R1.26 | Section 4.1 (marked PDF p. 8, margin l. 314) |
| R1.27 | Theorem 4.2 (marked PDF p. 9, margin l. 335) |
| R1.28 | Theorem 5.3 (marked PDF p. 11, margin l. 424) |
| R1.29 | Theorem 5.4 (marked PDF p. 11, margin l. 443) |
| R1.30 | Theorem 5.6 (marked PDF p. 12, margin l. 464) |
| R1.31 | Section 8 (marked PDF p. 16, margin l. 647); Appendix B (marked PDF p. 24, margin l. 943) |
| R1.32 | Section 8 (marked PDF p. 16, margin l. 647); Appendices A–B (marked PDF p. 22, margin l. 870) |
| R1.33 | Theorem 9.2 and Section 9.2 (marked PDF p. 20, margin l. 788) |
| R1.34 | Section 9 (marked PDF p. 18, margin l. 728) |
| R1.35 | Proposition 9.1 (marked PDF p. 19, margin l. 753); Sections 9.2–9.3 (marked PDF p. 20, margin l. 780) |
| R1.36 | Lemma 9.6 (marked PDF p. 21, margin l. 814) |
| R1.37 | Proposition 9.7 (marked PDF p. 21, margin l. 829) |
| R2.1 | Introduction (marked PDF p. 1, margin l. 23); Section 3 (marked PDF p. 5, margin l. 177); Proposition 6.3 (marked PDF p. 13, margin l. 524); Lemma 7.6 (marked PDF p. 16, margin l. 630); Sections 8–9.2 (marked PDF p. 16, margin l. 647) |
| R2.2 | Section 4.1 (marked PDF p. 8, margin l. 314); Sections 5.1–5.2 (marked PDF p. 9, margin l. 365) |
| R2.3 | Theorem 8.3 (marked PDF p. 17, margin l. 668); Proposition 8.4 (marked PDF p. 17, margin l. 687) |
| R2.4 | Sections 4–5 (marked PDF p. 8, margin l. 313); Appendices A–B (marked PDF p. 22, margin l. 870) |
| R2.5 | Proposition 3.1 (marked PDF p. 5, margin l. 183) |
| R2.6 | Section 5.1 (marked PDF p. 9, margin l. 365) |
| R2.7 | Section 5.1 (marked PDF p. 9, margin l. 365) |
| R2.8 | Section 5.1 (marked PDF p. 9, margin l. 365); Theorem 5.3 (marked PDF p. 11, margin l. 424) |
| R2.9 | Theorems 5.3–5.4 (marked PDF p. 11, margin l. 424) |
| R2.10 | Corollary 5.5 (marked PDF p. 12, margin l. 449) |
| R2.11 | Proposition 8.1 (marked PDF p. 16, margin l. 651); Lemma 8.2 (marked PDF p. 17, margin l. 658); Appendix B (marked PDF p. 24, margin l. 943) |

## Response to the Editor

**E.1. Editor (quoted excerpt):** "The reviewers agree that the scientific contribution has merit. However, they both point to many issues with the presentation, many of which leak also into the scientific part of the paper and are not merely language revisions. This is especially relevant in the learning part of the paper. ... Specifically, any correctness issues should be carefully considered."

**Response:** We agree. The submitted learner has been replaced by a conservative Gold learner; the terminal-rule gap has been repaired in both the witness set and the completeness proof; and the unused outer-context annotations have been removed. Fixed-$h$ substitutability is now stated for nonempty internal fragments, with $\lambda$ handled separately. Section 4 defines the batch operator and sequential learner before Section 5 introduces target-side yield typing for completeness. Polynomial characteristic data are attributed to the set-driven batch operator, while the sequential learner is asserted to have Gold stabilization with polynomial-time updates.

**Revision location:** Definition 2.1; Section 4.2; Section 5.1 and Theorem 5.3; Corollary 5.5; Sections 6–8.

## Reviewer 1

### General assessment and learning algorithm

**R1.1. Comment (quoted excerpt):** "The main result of this submission is an elegant generalization of the results of Clark and Eyraud (2007) on the learning of substitutable context-free languages (SCFLs) and of Yoshinaka (2008) on the learning of k,l-SCFLs. ... The only serious error is in the definition of their learning algorithm, but this is easily repairable by following prior work."

**Response:** We thank the reviewer for identifying the central correctness problem. Recomputing the batch grammar after every datum need not stabilize syntactically. The revision therefore separates a set-driven operator $\mathcal{B}_h$ (Section 4.1) from a conservative sequential learner $\mathcal{A}_h$ (Section 4.2), using the same conservative rebuild-on-missed-positive-example pattern as Clark–Eyraud's SGL and Yoshinaka's $(k, \ell)$-SGL. Corollary 5.5 proves eventual syntactic stabilization. We also now cite Eyraud–Heinz–Yoshinaka (2016, Appendix) at the point where their delayed-rebuild phenomenon is relevant: a characteristic sample may already have been seen while an earlier hypothesis persists, but the first later positive example that it misses forces a reconstruction from the accumulated data. Polynomial characteristic-data claims are made for the batch operator; the sequential learner is claimed only to have polynomial-time updates together with Gold convergence. In particular, $\mathcal{A}_h$ is not claimed to satisfy the incremental polynomial-time-and-data criterion of Eyraud–Heinz–Yoshinaka (2016, Definition 2.5): it may rebuild only after a characteristic sample has already appeared. Our polynomial characteristic-data guarantee for $\mathcal{B}_h$ is set-based (ibid., Definition 2.8), and applies only in the stated quantitative regimes, not uniformly for all $h$. Section 2.1 of the revised manuscript now explicitly makes this distinction.

**Revision location:** Section 2.1; Section 4.2; Corollary 5.5; Theorem 7.2; Theorem 8.3 and the following clarification.

**R1.2. Comment (quoted excerpt):** "On the other hand, the presentation leaves considerable room for improvement. By addressing the absence of definitions for technical terms, unnecessarily long proofs, inconsistent use of metavariables, and the over-fragmentation of the paper into too many sections, the authors could substantially reduce the burden on the reader, and I recommend a major revision along these lines. ... I suspect that this material could be presented in less than half the current length."

**Response:** We have substantially shortened the presentation. The submitted manuscript compiled to 40 pages; the revised additions-marked manuscript is 28 pages, including appendices and references. The 28-page clean revision likewise places the technical appendices after the main text. We removed repeated explanatory passages and shortened the Introduction, Conclusion, and linear normalization without discarding proof obligations. The locator guide gives revised statement anchors together with verified page and margin-line numbers from the compiled additions-marked manuscript. Numbered theorem-like environments have been reduced by more than half, main-text sections from 11 to 10, and main-text subsections from 31 to 16. Section 2 consolidates definitions and the positive-data model, the former complexity section has been folded into Section 5, and the three boundary sections have been consolidated as Section 9. Sections 6–8 organize the quantitative discussion rather than preserving the old layout.

The revision also contains substantive material added in direct response to the reports. Proposition 3.5, added in the revision (the submitted Proposition 3.5 is now Proposition 3.3), expressly restates the *classical* locally trivial-semigroup criterion (Pin 2025, Propositions XI.4.17 and XIV.1.20) in positive-image/kernel notation. We also cite Pin (1997, Proposition 5.18) for the classical *generalized definite* terminology. It is not claimed as an original algebraic characterization. The grammar-thickness terminology is attributed to Wakatsuki–Tomita (1993) and Yoshinaka (2008), with the later exposition cited at Eyraud–Heinz–Yoshinaka (2016, Definition 2.11); Definition 2.12 is used only for the identification-in-polynomial-time-and-thick-data (IPTtD) learning criterion. The learning-specific result relevant to R2.1 is Corollary 7.5: it gives polynomial encoded characteristic data in ordinary grammar thickness when the positive image is a homomorphic image of a subsemigroup of a product of a finite locally trivial semigroup and a finite group. Lemma 7.4 now establishes, by lifting letter images through the finite product divisor, that this algebraic condition is equivalent to the fixed window–group kernel-refinement hypothesis of Theorem 7.2. Corollary 7.5 then applies that theorem and the thickness-preserving normalization to obtain the polynomial bound; no necessity claim is made for polynomial data bounds among all fixed typings. Proposition 3.5 separately records the classical exact criterion for fixed-window-determined typings. The

corresponding union equality with Yoshinaka's window classes is a class-level consequence of that older criterion. Proposition 8.4 supplies the nonregular linear example requested in R2.3; Proposition 9.7 answers R1.37 by placing $\mathsf{RS} \cap \mathrm{CFL}$ inside Clark's congruential family. Section 6 states the quantitative scope more precisely. The submitted manuscript claimed exact reconstruction, Gold identification, and polynomial-time hypothesis updates for general fixed-$h$ context-free targets, but explicitly did not claim a polynomial characteristic-data bound for that full class. The revision adds a parameterized characteristic-data bound for the set-driven operator $\mathcal{B}_h$, polynomial in source grammar size and the target-side typed-thickness parameter. This does not assert a polynomial bound in ordinary grammar thickness alone for arbitrary fixed $h$, nor does it attribute a polynomial-data guarantee to the conservative sequential learner. Sections 7–8 identify restricted settings in which ordinary thickness or grammar size suffices.

Undefined and unnecessary terminology is removed, the explanatory SSBNF footnote is incorporated into the main text, and metavariables are clarified: in distribution and reconstruction arguments $u, v$ denote contexts and $x, y$ denote fragments; the fixed-window definition now follows the same convention; the locally-trivial-semigroup proof now uses a finite-semigroup quotient argument in place of the former exponent-based counterexample; and the finite-state control construction uses $\mu_L, \mu_R$ for monoid registers. The counter alphabet uses $\uparrow, \downarrow, \#$ rather than the letter $u$ and a semicolon. Appendix A contains the window–finite-group typed-yield bound and thickness-preserving normalization. Appendix B contains the longer linear-spine normalization proof, while the short proof of Lemma 8.2 appears in Section 8. Appendix C contains the new substitutability proof for the bracketed doubling family. The linear normalization was moved or condensed as both reviewers suggested; Section 9 is shorter overall.

**Revision location:** Sections 1–2; Sections 4–9; Appendices A–C.

**Section 1**

**R1.3. Comment (quoted excerpt):** "Stating the paper's place within the field and the significance of the language families considered would make the paper more compelling. Conversely, the current text is too heavily weighted toward technical content and could be made more concise."

**Response:** Agreed. Section 1.2 now begins with a concise account of the fixed-$h$ batch reconstruction operator and conservative Gold learner as the learning foundation, expressly acknowledging that qualitative learnability also follows from Yoshinaka's general many-sorted signature framework. Rather than listing that qualitative existence as a separate novelty claim, the Introduction emphasizes two main contributions: (1) quantitative characteristic-data results, including representation sensitivity and the window–finite-group and linear bounds; and (2) structural and expressive results, including retyping closure, the classical fixed-window criterion, examples beyond fixed windows, counter/Dyck obstructions, and Clark's congruential family. The fixed morphism is presented as a finite comparison/typing bias, and the fixed-$h$ learning claim is explicitly separated from the expressive union $\mathsf{RS}$.

The historical account explicitly distinguishes the author's 2014 preprint from Yoshinaka's subsequent general many-sorted signature learning theorem. In particular, the fibers $h^{-1}(m) \cap \Sigma^+$ come from the homomorphism fixed in this paper; Yoshinaka's framework provides the general sorts-and-operations formalism. We separately identify which quantitative and algebraic results are particular to fixed finite-monoid typing.

**Revision location:** Section 1.

**R1.4. Comment (quoted excerpt):** "The notations $\mathcal{D}_L$ and $\mathcal{C}_h^{\mathrm{cf}}$ are not standard and require definitions. In particular, the phrasing 'the general context-free class $\mathcal{C}_h^{\mathrm{cf}}$' is liable to be misunderstood."

**Response:** Agreed. $\mathcal{D}_L(x)$ is defined at its first occurrence in the Introduction and again formally in the preliminaries. The notation $\mathcal{C}_h^{\mathrm{cf}}$ is now defined at its first use in the Introduction as the class of context-free $\sim_h$-substitutable languages. The unrestricted fixed-typing class $\mathsf{RS}_h$ is then introduced before $\mathsf{RS} = \bigcup_h \mathsf{RS}_h$. Section 3 develops these classes and $\mathcal{C}_h^{\mathrm{lin}}$ formally. The phrase "general context-free class" has been removed.

**Revision location:** Introduction; Section 2.2 and the opening of Section 3.

**R1.5. Comment (quoted excerpt):** "The contributions are enumerated as (1)-(6), but it is unnecessary to count technical details as separate contributions. Moreover, at this point the text uses many terms whose technical meaning has not yet been made clear, and readers may not be able to follow them."

**Response:** Agreed. Instead of the submitted six-item list, Section 1.2 now gives a short, unnumbered paragraph describing the qualitative learning construction and its relationship to Yoshinaka's general framework, followed by just two numbered contributions: quantitative guarantees and structural/expressive position. Proof-local devices are not counted as separate contributions. The quantitative item describes shortest yields before and after splitting target nonterminals by their $h$-values; the formal typed-thickness terminology is introduced later in Section 6. The structural item distinguishes retyping closure from closure of a single fixed-$h$ slice and explicitly labels the window criterion classical; the formal definition of *locally trivial* remains in Section 2.2. An elementary

example $L = \{a, aa\}$ now follows the fixed-$h$ definition in Section 1.1: the trivial typing fails because two factors share a context but have different distributions, while a parity typing separates them. This illustrates the condition; it is not a new separation from fixed-window classes. The Introduction also defines $\mathsf{RS}_h$ before the expressive union $\mathsf{RS}$, separating fixed-$h$ learning from varying-typing comparisons.

**Revision location:** Section 1.1 (minimal example after the definition of fixed-$h$ substitutability); Section 1.2.

**R1.6. Comment (quoted excerpt):** " 'Yoshinaka's classical language' is often called 'the Łukasiewicz language' in the literature."

**Response:** Agreed. We use "Łukasiewicz language," add the Autebert–Berstel–Boasson (1997) handbook reference, and state its relation to the one-bracket Dyck language concisely. We also now cite Yoshinaka (2008, Section 3) for the prior observation that this language is outside every fixed $(k, \ell)$ class, and distinguish our stronger conclusion that it lies outside the finite-typing union $\mathsf{RS}$.

**Revision location:** Section 9.3.

**Section 2**

**R1.7. Comment (quoted excerpt):** "I doubt the claim that 'the learners considered in this paper produce their output independently of the order in which positive examples are observed.' It appears that the learning algorithm is being conflated with the grammar-construction algorithm that it uses as a subroutine. Indeed, the 'learning algorithm' in Section 5 does not converge, and hence does not satisfy Gold's criterion. Even if the learning algorithm were corrected to follow Clark & Eyraud, the output would still differ, when learning, e.g., the language $a^n c b^n$, between the case where the characteristic sample itself is received first and the case where the first positive example received is very long."

**Response:** We agree. We withdrew the submitted order-independence claim and distinguished the set-driven batch operator from the conservative sequential learner. Section 4.2 explains the presentation-order dependence: placing a sufficiently long positive example early in a text may retain an extra observed nonterminal in the stabilized grammar. Thus two texts can yield different final grammars for the same target, as in the reviewer's center-marked example. The revised text also notes that Clark (2013, Section 4, Example 1) exhibits the same presentation-order phenomenon for $L_c = \{a^n c b^n : n \geq 0\}$, with different stabilized substring-indexed grammars for different initial positive examples.

**Revision location:** Section 2.1; Section 4.2; Corollary 5.5.

**R1.8. Comment (quoted excerpt):** "This section should thoroughly define the necessary notation and terminology, including basic notions such as (linear) context-free grammars - some terms appear to be defined ad hoc in footnotes. Those terms include 'type', 'reconstruction', 'realize', 'reachable', 'observation', 'control', 'local configuration', etc. If these words carry no mathematical definition required to read the paper and were introduced only to aid intuition, then in the present form they tend to confuse the reader, and in some cases it would be better not to name them at all. A flood of excessive terminology can itself burden the reader and impede smooth understanding."

**Response:** Agreed. Section 2 now consolidates the grammar and learning notions needed later, including linear CFGs, reachable/productive/reduced grammars (trimming removes nonproductive symbols before unreachable ones), the positive-data model, set-driven reconstruction, characteristic samples, distributions, and the fixed-$h$ classes. Proof-local labels that did not carry mathematical content have been removed or are introduced only at their point of use; in particular, the former *finite typed reconstruction basis* terminology is gone. We also removed several notation collisions that impeded reading. Section 4.1 now uses the standard substring-indexed factor nonterminals $[x]$, with the bracket notation defined at first use. The revision also defines the letter-count notation $|w|_a$ in Section 2, spells out Clark–Eyraud's SGL as *Substitution Graph Learner* and Yoshinaka's $(k, \ell)$-SGL as $(k, \ell)$-*Substitutable Grammar Learner* at their point of use, specifies $\mathsf{Acc} \subseteq M$ at its first appearance, and removes the unused class notation $[x]_{\equiv_L}$. More specifically, *reachable* and yield *type* are defined in the preliminaries and typing setup, respectively; *reconstruction* refers to the defined batch operator $\mathcal{B}_h$ and its operation. *Observation* and *observed nonterminal* are not introduced as separate technical notions; Section 4.1 simply speaks of observed factors and their substring-indexed hypothesis symbols. *Realize* and *local configuration* have been removed, while *control* remains in standard uses such as control sets and pushdown finite control. These are presentational changes.

**Revision location:** Section 2; Section 4.1; Section 5.1; Appendix A.2.

**R1.9. Comment (quoted excerpt):** "The term 'finite typed reconstruction basis', a combination of undefined words, is clearly defined here; however, if it denotes an atomic concept for which none of its four constituent words is individually defined, a shorter term would be easier to understand."

**Response:** Agreed. Rather than rename this proof-local object, we removed both the term and the basis itself. The revised completeness argument uses only the yield-typed target grammar and the finite witness set $\mathsf{W}(\widetilde{G})$

that is actually needed for exact reconstruction.

**Revision location:** Section 5.1.

**Section 3**

**R1.10. Comment (quoted excerpt):** "The paper emphasizes (somewhat excessively and awkwardly) that h is fixed. To match this, RS_h should be defined before RS. The family RS is not shown to be learnable from positive examples (and indeed it is not learnable); what is learnable is RS_h."

**Response:** Agreed. The Introduction now explicitly defines $\mathsf{RS}_h$ before $\mathsf{RS} := \bigcup_h \mathsf{RS}_h$, and the Gold-identification theorem covers every *nonempty* language in $\mathcal{C}_h^{\mathrm{cf}} = \mathsf{RS}_h \cap \mathrm{CFL}$ for fixed $h$, not the whole class $\mathsf{RS}_h$; no learnability claim is made for the varying-$h$ union $\mathsf{RS}$, which is used only for expressive comparison. The revision also states that neither $\mathsf{RS}$ nor $\mathsf{RS} \cap \mathrm{CFL}$ is identifiable from positive data on a nonempty alphabet, since these classes contain all finite languages and $\Sigma^*$. We also reduced redundant uses of "fixed" in the Abstract and Introduction, while retaining the qualifier where it distinguishes the learnable slice from the varying-$h$ expressive union.

**Revision location:** Abstract; Introduction; Section 2.2 and the opening of Section 3.

**R1.11. Comment (quoted excerpt):** "Definition 3.2 requires care. On this point, the way Clark & Eyraud (2007) state their definition is somewhat careless; however, given that in Section 6.1 of that paper they state that $a^+$ is substitutable, and that their grammar-construction algorithm excludes the empty string, it is reasonable to interpret the intent of their definition as Yoshinaka's (2008) 0,0-substitutability. . . . "

**Response:** We agree with the reviewer's interpretation of Clark–Eyraud (2007): with nonempty internal factors, their intended case coincides with Yoshinaka's $(0, 0)$ case (Section 3.1). We now explicitly cite Clark (2013), as suggested, and restrict fixed-$h$ substitutability to nonempty internal fragments and handle $\lambda$ separately at the start symbol. Section 3.1 explains the issue explicitly for $L = a^+$: $(a, \lambda)$ belongs to both $\mathcal{D}_L(\lambda)$ and $\mathcal{D}_L(a)$, whereas $(\lambda, \lambda)$ belongs only to $\mathcal{D}_L(a)$. Thus comparing the empty factor would exclude $a^+$. We distinguish Clark–Eyraud's Definition 5, which quantifies over general strings, from the nonempty-fragment convention clarified by Clark (2013). The center-marked language $L_{\mathrm{c}} = \{a^n c b^n : n \geq 0\}$ is used consistently as an illustration; no argument concerning it relies on comparing an empty internal factor. The $(k, \ell)$ definition now uses $u_1, u_2, v_1, v_2$ for outer contexts, $p, q$ for the fixed prefix–suffix windows, and $r, s$ for the variable interiors of the compared fragments $prq$ and $psq$.

**Revision location:** Definition 2.1; the Clark–Eyraud special case in Section 3.1; the fixed-window definition in Section 3.1.

**R1.12. Comment (quoted excerpt):** "Proposition 3.5. $\sqcup$ and $\mathrm{pref}_k$ are undefined. . . . The proof is very long. If the multiplication on $T$ (as a monoid) were given explicitly in advance, then one line would suffice to explain that $\tau$ is a monoid homomorphism. Rephrasing in terms of $h_{k,\ell}$ or $M_{k,\ell}$ is needlessly cumbersome."

**Response:** We agree that the fixed-window argument was too long and used unnecessary notation. The symbol $\sqcup$ has been removed, and prefix and suffix notation, including the short-word convention, is defined before use. Proposition 3.3 (submitted Proposition 3.5) now presents the multiplication on the finite image directly and gives a short homomorphism check. We have retained the names $h_{k,\ell}$ and $M_{k,\ell}$ for a specific reason: $h_{k,\ell}$ is used explicitly in Proposition 3.5, Lemma 7.1, Theorem 7.2, and Lemma 7.4, while $M_{k,\ell}$ names the finite image on which the multiplication is defined. Thus we follow the suggestion to simplify the proof, while retaining these two reusable names. Short words, on which $(k, \ell)$-substitutability imposes no comparison, remain individually typed. The fixed-window counterexample criterion following Proposition 3.3 summarizes the separation argument used later. The classical locally trivial-semigroup criterion, the window–finite-group factorization, and the resulting thick-data bound are stated separately in Proposition 3.5, Lemma 7.4, and Corollary 7.5.

**Revision location:** Proposition 3.3 and the fixed-window counterexample criterion; Proposition 3.5; Lemma 7.4; Corollary 7.5.

**R1.13. Comment (quoted excerpt):** "Rather than saying that $h$ is given as data (the meaning is not crystal clear), it would be cleaner to state early in the paper something like 'fix an efficiently computable $h : \Sigma^* \to M$'. Define 'linear context-free'. . . . 'Explicit finite monoid homomorphism' is not defined. . . . Technically unnecessary words should be removed, especially in the statements of theorems."

**Response:** Agreed. We now fix a finite monoid by its multiplication table and letter images, explicitly state how $h(w)$ is computed, define linear CFGs in the preliminaries, and use "fixed efficiently computable homomorphism" in the main theorem.

**Revision location:** Sections 2–3.2.

**R1.14. Comment (quoted excerpt):** "I do not understand 'The proof for fixed $h$'. Do you mean 'The proof for Theorem 3.7'?"

**Response:** Agreed. This stray phrase has been deleted. The revised text now states the relevant theorem and its proof directly, without this wording.

**Revision location:** Section 3.2.

**Section 4**

**R1.15. Comment (quoted excerpt):** "Definition 4.2. 'productive' and 'reachable' are undefined. The phrasing 'This section proceeds in the order typed local control → finite typed basis.' is too casual."

**Response:** Agreed. These notions are defined in the preliminaries. The revised typed grammar is explicitly the reduced part of the full refinement, so thereafter all of its symbols are reachable and productive by convention and the repeated qualifications are removed. The quoted informal sentence has also been removed.

**Revision location:** Section 2; Section 5.1.

**R1.16. Comment (quoted excerpt):** "Footnote 2: 'productive' is already used earlier. Define it in Section 2 rather than in a footnote."

**Response:** Agreed. Reachable and productive are now defined in Section 2 before the target refinement is introduced, and the former explanatory footnote has been removed.

**Revision location:** Section 2; Section 5.1.

**R1.17. Comment (quoted excerpt):** "The text says 'the set of reachable and productive typed nonterminals'; is there any nonterminal of $\tilde{G}$ that fails this condition? This may be intended as a helpful reminder or emphasis, but it confuses the reader instead."

**Response:** The reviewer is correct: every symbol retained in $\widetilde{G}$ is reachable and productive because $\widetilde{G}$ is defined as the reduced part of the full yield-typed refinement. The repeated qualification has therefore been removed.

**Revision location:** Section 5.1.

**R1.18. Comment (quoted excerpt):** "The first paragraph states no reason why 'different occurrences must be treated as different typed copies' at all. Expressions making clear 'for what purpose and for what reason' are needed. Example 4.3 is labeled '(why typing is needed)', yet why it is needed is not stated. ... If the authors wish to explain the reason, the development would be smoother if they first presented the learning algorithm and then, within its completeness proof, stated: 'here we assume the target grammar is type-refined, and we show that the learner's hypothesis grammar can mimic it.' "

**Response:** We have adopted the reviewer's suggested order of presentation. Section 4 first defines the batch reconstruction operator $\mathcal{B}_h$ and the conservative sequential learner $\mathcal{A}_h$, with no target refinement assumed. Section 5 then introduces the yield-typed target grammar specifically for the completeness proof and states explicitly that the purpose is to show that the already-defined hypothesis grammar can mimic typed target derivations. Outer-context typing has been removed. The remaining yield typing is visible at the two unary Rule (U) steps in that simulation. For a typed terminal rule $X \to a$, it gives $h(\omega(X)) = h(a)$; for a typed binary rule, Rule (U) moves from the canonical yield $\omega(X)$ to $\omega(Y)\omega(Z)$, and the typed refinement gives $h(\omega(X)) = h(\omega(Y))h(\omega(Z))$. No outer-context annotation is used.

**Revision location:** Section 4; Section 5.1 and Theorem 5.3.

**R1.19. Comment (quoted excerpt):** "Lemma 4.4. It is confusing that an expression concerning derivations in $G$ appears immediately after 'In the full typed refinement $\widetilde{G}^{\text{full}}$,'. It is unnecessary or should be moved to the end of the sentence."

**Response:** The former Lemma 4.4 has been removed in the reorganization. Its needed content is now folded into Proposition 5.2, which states language preservation and the yield invariant directly and keeps derivations of the original grammar and of the typed refinement syntactically separate.

**Revision location:** Proposition 5.2.

**R1.20. Comment (quoted excerpt):** "Lemma 4.8. 'If $X$ is reachable and productive in $\widetilde{G}$,' This likewise confuses the reader."

**Response:** This is resolved by the same convention: $\widetilde{G}$ is the reduced part of the full yield refinement. The former Lemma 4.8 is deleted, together with the redundant reachability/productivity hypothesis.

**Revision location:** Section 5.1.

**R1.21. Comment (quoted excerpt):** "The opening paragraph (the definitions of $\omega$ and $\chi$) would connect better if placed immediately before the definition of CS."

"Since the term 'characteristic sample' is defined in Def. 2.1, avoid writing 'Define the characteristic sample by' here unless it is immediately clear that this set $CS(G)$ satisfies Def. 2.1."

"The term 'realizable' is defined here but used earlier, whereas the term 'realized', used in many places, is undefined."

"The phrase 'canonical observation word' could be defined when $CS(\widetilde{G})$ is defined."

**Response:** Agreed. The canonical yield $\omega(X)$ is defined after the trimmed refinement, before first use. The canonical context $\chi(X)$ appears immediately before the finite witness set. A short parity example now explains the need for yield typing in the main text. The set $\mathsf{W}(\widetilde{G})$ is called a witness set until exact reconstruction has been proved; only then is it identified as a characteristic sample. The unused basis, $NT(\cdot)$, Rule$(\cdot)$, and the realized/realizable and exhibited terminology have been removed.

**Revision location:** Section 5.1.

**R1.22. Comment (quoted excerpt):** "Lemma 4.6. This is far too trivial a corollary. Either label it as a Corollary or delete it."

"Lemma 4.7. Given what the reader has understood so far, this is trivial and needs no proof. . . . "

"Theorem 4.9. I cannot understand what this theorem shows. . . . "

"Theorem 4.12. The first sentence should be removed. The second and third sentences are not understandable owing to the undefined term 'exhibited'."

**Response:** The former Lemmas 4.6–4.7 and Theorems 4.9 and 4.12 have been removed. The typed-refinement development now consists of one normal-form definition, one invariant proposition (Proposition 5.2), and the witness-set construction.

**Revision location:** Section 5.1.

**R1.23. Comment (quoted excerpt):** "$\alpha$ is a string, not a symbol."

"Said 'In particular, if $X \to YZ$,' but the antecedent of the 'if' is unrelated to its consequent."

"Lemma 4.5. 'For (iii), every derivation in $G$ erases typed annotations' I cannot understand this. Do you mean 'For (iii), every derivation in $G$ with typed annotations erased gives . . . '?"

"Is $NT(\widetilde{G})$ not the same as $\widetilde{V}$ on page 9?"

"Definition 4.11. Is Rule$(\widetilde{G})$ different from $\widetilde{P}$ on page 9? . . . "

**Response:** Agreed. These sentences and duplicate notations disappear in the rewrite. The lifting statement is now Proposition 5.2, stated directly as language preservation plus the yield invariant; the proof says that erasing type annotations maps a typed derivation to a derivation of the original grammar. The former trivial corollaries are deleted (see also R1.17 for the reduced-refinement convention).

**Revision location:** Section 5.1.

**Section 5**

**R1.24. Comment (quoted excerpt):** "The learning algorithm as defined here does not converge. This is the fatal error of the paper."

**Response:** We agree. The submitted learning algorithm did not converge, and this was a central correctness error. As detailed in R1.1, we replaced it with a conservative sequential learner and proved syntactic stabilization in Corollary 5.5.

**Revision location:** Section 4.2; Corollary 5.5.

**R1.25. Comment (quoted excerpt):** "Corollary 5.8. This cannot be accepted as a proof. By the definition in Section 5.1, the learner's hypothesis grammar keeps growing as it receives more positive examples."

**Response:** See R1.1 and R1.24. We changed the learner rather than only the proof; Corollary 5.5 is the resulting stabilization statement.

**Revision location:** Section 4.2; Corollary 5.5.

**R1.26. Comment (quoted excerpt):** "Remark 5.1. I cannot grasp the point. What does 'soundness of a rule' mean? . . . If one sentence is to be added, something like 'the grammar is designed so that the language derived by each nonterminal $[x : u, v]$ should be the set of strings substitutable with $x$' might aid intuition. Stating a mathematical property concisely helps the reader more than deploying adjectives that carry no mathematical definition."

"Lemmas 5.3 and 5.4 merely repeat definitions and should be deleted."

**Response:** The remark and definition-restating lemmas are deleted. We also simplified the reconstruction further than the submitted notation: the hypothesis now has one symbol $[x]$ for each observed nonempty factor, rather than a separate symbol for each occurrence context. The mathematical property is now stated directly:

whenever $[x] \Rightarrow^* w$, the derived word satisfies $x \equiv_L w$ and $h(w) = h(x)$. We do not claim that a finite-sample nonterminal generates the entire syntactic-congruence class. The four displayed rule families are branching, $h$-filtered unary substitution, lexical emission, and start entry. The revised order is now: batch grammar, sample consistency and soundness, sequential learner, then target-side yield refinement and completeness.

**Revision location:** Section 4 (Rules (B), (U), (L), (S) and the soundness invariant).

**R1.27. Comment (quoted excerpt):** "Lemma 5.2. In the second sentence, the first claim is trivial and the second is not understandable. What is the corresponding local configuration? Its correspondence with the part following 'More precisely' is unclear."

**Response:** Agreed. The former Lemma 5.2 and the undefined "corresponding local configuration" terminology have been removed. Their role is replaced in the proof of Theorem 4.2 by the simpler invariant

$$[x] \Rightarrow^* w \implies x \equiv_L w \text{ and } h(w) = h(x),$$

proved by induction on derivation-tree height. The unary case uses one shared sample context and fixed-$h$ substitutability; the branching case uses that $\equiv_L$ is a monoid congruence. Start rules then give hypothesis soundness.

**Revision location:** Theorem 4.2 and its displayed soundness invariant.

**R1.28. Comment (quoted excerpt):** "Page 14, line 1: 'Since $G$ is in SSBNF' $\to \widetilde{G}$?"

**Response:** Corrected during the rewrite; the relevant grammar is named explicitly throughout the completeness proof.

**Revision location:** Theorem 5.3.

**R1.29. Comment (quoted excerpt):** "Theorem 5.7. Make the statement mathematical. What do you mean by 'a finite sample exposes the finite typed reconstruction basis'? A sample is a string set and the FTRB is a tuple."

**Response:** Theorem 5.4 now states the exact implication $\mathsf{W}(\widetilde{G}) \subseteq K \subseteq L \Rightarrow L(\mathcal{B}_h(K)) = L$ with all assumptions in the theorem statement.

**Revision location:** Theorem 5.4.

### Section 6

**R1.30. Comment (quoted excerpt):** "If the paper adopts the stance of Remark 6.4, then the fact that the grammar construction can be carried out in polynomial time is almost trivial, so the content of this section could be presented in a single paragraph."

**Response:** The former standalone complexity section is now a short subsection in Section 5.3, centered on one counting argument with $n_K = \sum_{w \in K}(|w| + 1)$: $O(n_K^2)$ factor nonterminals, $O(n_K^3)$ branching candidates, and $O(n_K^3)$ unary candidates by pairing each observed factor occurrence with each sample word and checking the corresponding context and $h$-type. Even when factor strings are copied literally, this gives a conservative explicit worst-case bound $O(n_K^4)$. The update consequence is now stated directly in prose, rather than as a separate corollary, followed by a brief limitation example. The size convention also counts $\lambda$ correctly.

**Revision location:** Section 5.3.

### Section 7

**R1.31. Comment (quoted excerpt):** "This is a special case of the arguments in the preceding sections and could be much shorter ... Too long explanation may blur the focus — what the reader should understand, and what differs from the preceding sections."

"Lemma 7.3. 'strict linear derivation' is not defined."

**Response:** We substantially shortened the material corresponding to submitted Section 7, now Section 8. The main text retains only Proposition 8.1 and Lemma 8.2, which provide the single-spine form and short-yield bound used by Theorem 8.3; the short proof of Lemma 8.2 follows its statement in Section 8, while the longer normalization proof remains in Appendix B, where the binarization cases are expressed as two uniform chains. The undefined term *strict linear derivation* has been removed.

**Revision location:** Proposition 8.1 and the surrounding discussion; Lemma 8.2; Appendix B.

**R1.32. Comment (quoted excerpt):** "Proposition 7.2 does not need two full pages on the proof. I do not know whether exactly the same restriction has been defined before, but it is a combination of existing restrictions that are known to be inessential, and most readers will accept Proposition 7.2 as trivial or as folklore."

"Lemmas 7.5 and 7.6 are too trivial."

**Response:** Agreed. The former two-page normalization proof and the elementary intermediate lemmas have been removed from the main text. Only Proposition 8.1 and Lemma 8.2 remain as the intermediate statements, with the brief short-yield proof given directly after Lemma 8.2 and the longer normalization proof in Appendix B; the former trivial lemmas are absorbed into those arguments or deleted.

**Revision location:** Proposition 8.1 and the surrounding discussion; Lemma 8.2; Theorem 8.3 and the following clarification; Appendix B.

**Sections 8–10**

**R1.33. Comment (quoted excerpt):** "The choice of the alphabet $\{d, u, ;\}$ is unfortunate. The semicolon in particular is very hard to read within the running text, and $u$ is a source of confusion because it almost always denotes a string elsewhere in the paper. . . . "

"The proof of Theorem 8.6 should be simplified. Introducing complex formulas and lavishing Greek letters in an attempt to shorten the witness is not the essence of the theorem and only burdens the reader. . . . "

**Response:** Agreed. We adopted the reviewer's suggested witness in its essential form, renaming $d, u, ;$ as $\uparrow, \downarrow, \#$ and $p$ as $\rho$. Theorem 9.2 now uses $(\uparrow\downarrow)^k \uparrow^{\rho-1} \#^\ell$ and $(\uparrow\downarrow)^k \uparrow^\rho \#^\ell$. They share the empty context and the same fixed-window boundaries; an additional preceding $\uparrow$ distinguishes their distributions. The criterion following Proposition 3.3 avoids an expanded six-variable argument. We thank the reviewer for this simpler witness. The sharpness of the boundary $\rho \geq 2$ is now justified in a footnote: $\mathrm{CTR}_1$ is $(1, 1)$-substitutable because membership is determined by an initial-symbol condition and three forbidden adjacent pairs. For nonempty admissible factors with a common context, matching first and last symbols imply identical membership under every context.

**Revision location:** Section 9.2.

**R1.34. Comment (quoted excerpt):** "There is no need to devote an entire subsection to this material. It is unclear what Corollary 8.7 establishes. . . . Proposition 3.5 and Theorem 8.6 are meaningful; but folding them into Corollary 8.7 actually obscures the paper's important contributions."

" 'We note that a block of a word in CCL refers to each of the subwords delimited by the separator symbol ;.' This is a new definition, not a note. In fact, the proof of Theorem 8.6 uses 'block' of a CCL string in an entirely different sense."

"Sections 8-10 all concern the richness of the class of the learning target languages; consolidating them into a single section and keeping the explanations and proofs concise would help the reader grasp the overall picture."

**Response:** We agree that a union of separately learnable families need not itself be learnable. The former Corollary 8.7 has therefore been deleted: RS is used only for expressive comparison, whereas learning is asserted for fixed-$h$ context-free slices. The former three boundary sections are consolidated in Section 9. The overloaded *block* terminology from the capped-counter discussion is replaced by explicit factors and height-based descriptions; the capped family is written $\mathrm{CTR}_\rho$ over $\{\uparrow, \downarrow, \#\}$ and its witnessing $h_\rho$ is the transition-monoid morphism of the DFA described in Section 9.2. Yoshinaka's earlier regular separation (2008, Proposition 1) is cited briefly before Proposition 3.2 in Section 3; the explicit counterexample $L_0$ is now defined only in Section 9.2.

The submitted Section 10 example $\Delta^* = (\{a^n b^n : n \geq 0\})^*$ now appears in shorter form as Proposition 9.1, including its deterministic context-free justification. Its finite typing $h_\star$ records first/last symbols and the presence of $ba$. The proof uses the balance $\beta$ and the canonical-completion definition of the admissible entry-height sets $\mathcal{Q}(x)$. The concatenation identity for $\mathcal{Q}(st)$ has a short footnoted proof. If both factors contain $ba$, their $\mathcal{Q}$ sets are at most singletons with a common entry height; if both lie in $a^*b^*$, their sets coincide because $\beta(x) = \beta(y)$. This makes the substitutability argument explicit without auxiliary prefix-minimum notation. The separate Kleene-star nonclosure example based on $\{a^n b a^n : n \geq 0\}$ and the appendix on $\mathrm{CCL}_1$ have been deleted. The intersection and inverse-homomorphism results from the submitted Appendix A are condensed in Proposition 3.2, with the erasing case handled for the revised nonempty-fragment convention.

**Revision location:** Section 9.

**R1.35. Comment (quoted excerpt):** "Definition 9.1. Define the symbol #, and make its use consistent in Section 9."

"Proposition 9.2 is trivial and requires no proof. Proposition 9.6 is also a well-known fact, as already stated in Ginsburg & Greibach, Deterministic context-free languages, Information and Control 9 (1966), 620-648."

"Lemma 9.8. The relationship between the Łukasiewicz language and the Dyck language is also well known."

"The proof of Proposition 10.3 may also be omitted or greatly shortened. In the proof of Proposition 10.5, consider aligning the symbols with those of Definition 3.3."

**Response:** The reset symbol # is defined once and used consistently. The deterministic context-freeness of the uncapped counter CTR (former Proposition 9.2) is stated briefly via a deterministic pushdown counter with

reset. For $D_1$ (former Proposition 9.6) and the Łukasiewicz language, we cite Ginsburg–Greibach (1966); the identity $L_{\mathrm{Luk}} = D_1 b$ (former Lemma 9.8) is cited to Autebert–Berstel–Boasson (1997), and Yoshinaka (2008, Section 3) is credited for the fixed-window non-substitutability result. The quotient argument establishes the stronger exclusion from RS. The deterministic pushdown recognition argument from submitted Proposition 10.3 is condensed in Proposition 9.1. That proposition now uses a direct definition of $\mathcal{Q}$, seven simple examples, and a footnoted concatenation identity in the substitutability proof. The former Proposition 10.5 proof is replaced by the criterion following Proposition 3.3, and the capped-counter regularity argument is now one elementary DFA paragraph.

**Revision location:** Section 9.

**R1.36. Comment (quoted excerpt):** "Lemma 9.11 is a fundamental property of $\sim_h$ substitutability; since the basic mathematical properties of $\sim_h$ substitutability are collected in an appendix, placing it there is one option."

**Response:** The submitted appendix of basic properties is no longer retained as such: its intersection and inverse-homomorphism facts are condensed and adapted in Proposition 3.2. The remaining right-quotient property is now the two-line Lemma 9.6, used immediately in the following argument. We keep its brief proof in place rather than creating an appendix for this one fact.

**Revision location:** Lemma 9.6.

#### Section 11

**R1.37. Comment (quoted excerpt):** "It is worth mentioning that $\mathsf{RS} \cap \mathrm{CFL}$ is a subset of the class of Clark's congruential CFLs."

**Response:** Added as Proposition 9.7. Clark (2010) places substitutable and $(k, \ell)$-substitutable languages in the congruential family; the revised text also explicitly identifies his Dyck example. The proposition extends the containment slice-wise to every fixed $h$, hence to $\mathsf{RS}(\Gamma) \cap \mathrm{CFL}$. Properness over $\{a, b\}$ is witnessed by $D_1$, which is congruential but lies outside RS.

**Revision location:** Proposition 9.7.

## Reviewer 2

For the learning-algorithm correction, see R1.1 and R1.24; for the terminal-rule completeness repair raised below, see R2.8.

#### General positioning of fixed-$h$

**R2.1. Comment (quoted excerpt):** "At a general level, the paper offers an abstraction of the learning of Yoshinaka's $(k, \ell)$-substitutable languages by replacing the fixed-length window by an arbitrary recognizable congruence, and lifts the known results accordingly. The abstraction places the problem in a powerful and well-understood framework, but it requires a whole morphism to be known in advance for each language, whereas the earlier settings fixed a single uniform parameter valid across the entire class. The paper acknowledges that learning the morphism is outside its scope, yet the interest of the setting – beyond providing a more abstract framework for the study of $(k, \ell)$-substitutable languages – remains unclear. A promising direction would be to deepen the links with pushdown automata, which Sections 8 to 10 already touch on. The links with the work of Takada ... are also worth investigating. In a more practical direction, this work could be read as extending automaton typing ... to context-free substitutable languages, and the conditions a typing must satisfy could be studied."

**Response:** We agree that specifying a finite-monoid homomorphism requires a stronger a priori bias than choosing a pair $(k, \ell)$. Our learner does not infer this homomorphism. Instead, a single $h$ is fixed *uniformly* for the entire class $\mathcal{C}_h^{\mathrm{cf}}$, just as $(k, \ell)$ is fixed for Yoshinaka's class. We make no learnability claim for the union RS over varying homomorphisms. This uniform bias yields structural and quantitative consequences: retyping closure, the window–finite-group and linear characteristic-data bounds, and separation results beyond fixed windows. Moreover, when $h = h_{k,\ell}$, our reconstruction specializes to Yoshinaka's substring-rule scheme (2008, Section 4), up to the minor representational differences explained in revised Section 4.1.

The revised Introduction also situates this fixed-typing bias alongside Kanazawa's study of restricted learnable classes of categorial grammars (1996, 1998, Chapters 5 and 8). The constraints and learning techniques are different; no implication from Kanazawa's results to our fixed-$h$ theorem is claimed.

The revised Introduction also acknowledges Yoshinaka (2015, Theorem 1) as a general signature-based qualitative learning result. The nonempty fibers $h^{-1}(m) \cap \Sigma^+$ in that comparison are determined by our fixed finite-monoid homomorphism, and are represented as distinct sorts within Yoshinaka's general framework; they are not

attributed to Yoshinaka as his particular choice of types. The qualitative comparison is kept distinct from our finite-monoid-specific quantitative and structural results.

The finite-typing framework has structural and quantitative consequences. Section 3 now contrasts Yoshinaka's nonclosure results (2008, Proposition 1) with Proposition 3.2 directly, without reproducing his regular counterexample there. His inverse-image counterexample uses erasure, whereas nonerasing inverse images preserve fixed-window substitutability. Proposition 3.2 instead accommodates regular filtering and erasing inverse images by changing the finite typing, not by claiming closure of an unchanged fixed-window or fixed-$h$ slice. A new footnote to Proposition 3.2(iii) illustrates why the erasure flag $e_\varphi$ is needed: with $L = a^+$, trivial $h$, and $\varphi(a) = a$, $\varphi(b) = \lambda$, the inverse-image factors $a, b$ share a context and have the same $(h \circ \varphi)$-type, but different distributions. The known regular counterexample $L_0$ is defined inline in Section 9.2, where the capped-counter family is compared with the prior separation. Proposition 3.5 restates the classical locally trivial-semigroup criterion in the present positive-image setting. New Lemma 7.4 separates the semigroup-theoretic step: a letterwise lift into a locally trivial–group product proves the equivalence with window–group kernel refinement. Corollary 7.5 then gives the polynomial characteristic-data bound using the typed-yield estimate in Section 7 and Appendix A.1 and the grammar normalization. This algebraic reformulation is not claimed as a new semigroup theorem. This extends the ordinary-thickness characteristic-data bound without learning the typing $h$ or covering arbitrary fixed $h$. Proposition 8.4 exhibits a nonregular linear fixed-$h$ language outside every fixed-window class. The new Proposition 6.3 and Lemma 7.6 provide a quantitative counterpoint: the fixed two-element semilattice typing, outside $\mathbf{LI} \vee \mathbf{G}$, requires exponentially long data even though the same finite targets have polynomial data under the trivial typing. Proposition 6.3(iv) makes the characteristic-sample lower bound independent of the choice of a set-driven operator admitting characteristic samples for the distinguishing pair. Corollary 6.4 strengthens the typed-thickness lower bound to $3 \cdot 2^n - 2$ for any reduced SSBNF presentation. This does not assert necessity of the algebraic condition for every typing or a running-time or convergence-stage lower bound for arbitrary Gold learners. The classical algebraic criterion and previously known closure observations are not claimed as newly discovered in this revision.

The two-word example in Section 1.1 makes the basic type-separation mechanism concrete; it does not establish a separation from fixed-window classes. The latter distinction rests on the subsequent structural and expressive results. Section 3 briefly explains typing adequacy as separation of distributional conflicts by a finite-index monoid congruence. It also notes that requiring the typing kernel to refine the full syntactic congruence would force regularity. This describes the required bias, not an algorithm for finding it. Section 8 also gives a limited connection with Takada's regular control sets for linear grammars. We claim neither equivalence with Takada's learning framework nor a general pushdown characterization.

**Revision location:** Introduction; Section 3 (concise Yoshinaka comparison immediately before Proposition 3.2); Section 4; Propositions 3.2 and 3.5; Main Theorem 3.4(iv); Proposition 6.3, Corollary 6.4, and Appendix C; Section 7, Lemmas 7.4 and 7.6, Corollary 7.5, Appendix A.1; Proposition 8.4; Section 9.2 ($L_0$ definition); the control-set discussion in Section 8.

**Main concern 1: outer context typing and transport**

**R2.2. Comment (quoted excerpt):** "My main concern is the typed refinement, and specifically transport. Rule (2) allows a nonterminal to move between arbitrary contexts with no condition whatever on their types. Is this intended? If the outer components $m, n$ record where a nonterminal may appear, why does the rule that relocates it ignore them entirely? As far as I can tell the components are then read by no proof: Lemma 4.6 is invoked only for the yield component, and replacing $A_p^{m,n}$ by $A_p$ throughout leaves every result intact. Does the author agree, and if not, where exactly is an outer type consulted? The situation seems to me to follow from Definition 2.3 itself, whose algebraic condition bears on the substituted fragment and imposes nothing on the surrounding context – so that a learner mirroring that definition has nothing to guard transport with. If that is right, then the typed transport announced in Section 1.1 does not occur anywhere in the article, and the claim that occurrences must be separated according to their outer types cannot be established by any example. The yield typing, which is genuinely essential and is what becomes vacuous under the trivial monoid, should not be left in a footnote."

**Response:** We agree, and the observation permits a stronger simplification than merely deleting the outer type annotations. No proof in the *submitted version* consulted the outer components, so $A_p^{m,n}$ has been replaced by the yield-typed $A_p$ on the target side. On the hypothesis side, the context index is also inessential: submitted Rule (2) made all symbols $[x : u, v]$ with the same factor $x$ mutually derivable. Quotienting those symbols therefore gives one nonterminal $[x]$ per observed factor. Submitted Rule (3) then becomes exactly the unary rule $[x] \to [y]$ when $x$ and $y$ share an observed context and $h(x) = h(y)$. The separate transport rule disappears.

Apart from the start-symbol presentation, the treatment of $\lambda$, and redundant identity rules, the revised batch grammar follows Yoshinaka's substring-indexed rule scheme (2008, Section 4), also presented by Clark (2013,

Section 4). For window typing $h_{k,\ell}$ its nontrivial unary rules coincide with Yoshinaka's; for general fixed $h$ they require type equality and a shared sample context. Soundness now uses the invariant

$$[x] \Rightarrow^* w \quad \Longrightarrow \quad x \equiv_L w \text{ and } h(w) = h(x).$$

Yield typing remains necessary only for the present witness-based completeness argument: it licenses the unary steps from a canonical parent yield to a typed terminal yield or to the concatenation of typed child yields. The parity example is worded explicitly as an illustration of this proof requirement, not as an algorithm-independent necessity.

**Revision location:** Section 4; Section 5.1 and Theorem 5.3.

**Main concern 2: extent of the linear theorem**

**R2.3. Comment (quoted excerpt):** "The linear result raises a different question: how large is the class it covers? The three closing sections situate RS, not $\mathcal{C}_h^{\text{lin}}$, so the reader is given no sense of what Theorem 7.9 reaches beyond the linear substitutable languages. The capped-counter languages are regular and hence linear, so the separation from Yoshinaka's hierarchy already holds inside a linear class, whereas $\{a^n b^n\}^*$ is not linear, so the final section establishes nothing at that level. As it stands the linear theorem reads as a technical strengthening on a class of unknown extent, the non-regular linear case being left open."

**Response:** Section 8 now provides both a concrete nonregular linear example and a structural closure mechanism. Proposition 8.4 proves that $L_{\pm,e}$ is linear, nonregular, and substitutable under a fixed finite-monoid typing, but belongs neither to the Clark–Eyraud substitutable class nor to any $(k,\ell)$-substitutable class. Its main proof obtains the fixed typing from the regular-filter construction below; a subsequent short paragraph exhibits a concrete four-element typing.

More generally, Proposition 3.2(ii), together with closure of linear languages under regular intersection, gives

$$L \in \mathcal{C}_h^{\text{lin}}, \quad R = g^{-1}(F) \quad \Longrightarrow \quad L \cap R \in \mathcal{C}_{h\times g}^{\text{lin}}.$$

The proof of Proposition 8.4 uses the explicit instance $L_{\pm,e} = L_{\text{all}} \cap R$, where

$$L_{\text{all}} = \{a^n \xi b^n : n \geq 0,\ \xi \in \{c, d, e\}\}$$

is linear and Clark–Eyraud substitutable under the nonempty-factor convention, and

$$R = (aa)^* c (bb)^* \cup a(aa)^* d\, b(bb)^* \cup a^* e b^*.$$

The regular-filter argument proves membership in a fixed-typing linear class via Proposition 3.2(ii). The four-element typing is recorded separately as a more economical concrete witness, with a short verification. The regular-level separation was already known from Yoshinaka (2008, Proposition 1); the capped-counter family is retained only as the finite-state companion to the uncapped counter CTR. No separation from Takada's control-set approach is claimed.

We withdraw the submitted Section 7 title, *The Linear Subclass: A Full Polynomial Time-and-Data Theorem*, as a characterization of the sequential learner. Theorem 8.3 now establishes polynomial characteristic data only for the set-driven operator $\mathcal{B}_h$, relative to a chosen linear CFG representation of a fixed-$h$ substitutable target. This does not contradict de la Higuera's negative result for the unrestricted class of linear languages: our bound applies only to $\mathcal{C}_h^{\text{lin}}$ for fixed $h$. We also claim no incremental polynomial-data guarantee for the sequential learner $\mathcal{A}_h$.

**Revision location:** Proposition 3.2(ii); Theorem 8.3 and its scope clarification; Proposition 8.4 and the following regular-filter paragraph.

**Further comments**

**R2.4. Comment (quoted excerpt):** "The paper would gain considerably from being simplified: much of Section 4 is definitional, several results restate what their proofs have just established, and objects are regularly used before they are introduced, which forces the reader to hold material in suspense. Relatedly, since $\tilde{G}$ is by construction the trimmed refinement of $\tilde{G}_{\text{full}}$, its symbols are reachable and productive by definition; repeating this qualification at every occurrence adds length without adding content, and one sentence recording the convention would suffice."

**Response:** We agree. The repeated reachable/productive qualifications have been removed and the reduced-refinement convention is stated once in Section 5.1. The development is reordered so that the batch reconstruction operator is defined before the target-side refinement used in completeness. Within the refinement, the canonical

yield $\omega$ is defined before first use and the canonical context $\chi$ immediately before the witness set. The quantitative material is then separated into the general typed-thickness analysis of Section 6, the window–finite-group specialization of Section 7, and the linear specialization of Section 8; long normalization proofs are moved to the appendices. The later algebraic addition follows the same rule: the term *locally trivial* is defined formally in Section 2.2 before its use in Theorem 3.4.

**Revision location:** Section 4; Section 5.1; Sections 6–8; Appendices A–B.

**R2.5. Comment (quoted excerpt):** "Proposition 3.1: the hypothesis $L = h^{-1}(F)$ characterises the regular languages exactly. Replace 'In particular' by 'Thus' (and the proof could make explicit why)."

**Response:** Agreed. Proposition 3.1 states the finite-monoid recognition characterization directly and concludes that every regular language belongs to $\mathsf{RS}$. Its proof cites Eilenberg (1974, Chapter III, Propositions 10.1–10.2) for the standard characterization and spells out the needed consequence: if $L = h^{-1}(\mathsf{Acc})$ and $h(x) = h(y)$, then $h(uxv) = h(uyv)$ for all contexts $(u, v)$, so the distributions agree.

**Revision location:** Proposition 3.1.

**R2.6. Comment (quoted excerpt):** "Definition 4.11: the basis $B(\widetilde{G})$ is never used. Is it useful to introduce it as well as Theorem 4.12 and Remark 4.13? (Notations NT() and Rule() are unnecessary since it is sufficient to define $G = ()$ and use defined $V$ and $P$.)"

**Response:** Agreed. These objects and notations have been removed.

**Revision location:** Section 5.1.

**R2.7. Comment (quoted excerpt):** "Example 4.3 is not in SSBNF"

**Response:** The former example has been removed entirely; the revised proof no longer needs it.

**Revision location:** Section 5.1.

**R2.8. Comment (quoted excerpt):** "Lemma 5.2 (i): If $X$ carries several terminal rules $\omega(X)$ is not sufficient, and rule (3) should be used (proof needs to be corrected accordingly)."

**Response:** The reviewer is correct; this was a genuine gap in the submitted completeness argument. The witness set now contains $u_X a v_X$ for every typed terminal rule $X \to a$. Together with the canonical witness $u_X \omega(X) v_X$, this places $a$ and $\omega(X)$ in the same observed sample context. The typed refinement gives $h(\omega(X)) = h(a)$, so the terminal base case applies unary Rule (U) from $[\omega(X)]$ to $[a]$ when necessary, followed by lexical Rule (L).

**Revision location:** Section 5.1 (witness set); Theorem 5.3 (terminal-rule base case).

**R2.9. Comment (quoted excerpt):** "Are Lemmas 5.3 and 5.4 needed?"

**Response:** Agreed. They are deleted.

**Revision location:** Sections 4–5.1.

**R2.10. Comment (quoted excerpt):** "Corollary 5.8 Should begin with 'For a fixed homomorphism $h$ from $\Sigma^*$ to a finite monoid $M$, ...', or 'Being given a homomorphism $h$ from...' "

**Response:** The revised Gold-identification statement begins in the requested form: *For a fixed homomorphism* $h : \Sigma^* \to M$ *into a finite monoid* $M$, ....

**Revision location:** Corollary 5.5.

**R2.11. Comment (quoted excerpt):** "section 7.1 Transformation to standard SSLNF (3 pages) could be condensed or given in appendix."

**Response:** We took the appendix option. Of the normalization machinery, Section 8 retains only Proposition 8.1 and Lemma 8.2, the statements used directly by Theorem 8.3. The longer normalization proof remains in Appendix B, while the short proof of Lemma 8.2 is provided in Section 8 to make the polynomial-bound argument self-contained.

**Revision location:** Proposition 8.1 and the surrounding discussion; Lemma 8.2; Theorem 8.3 and the following clarification; Appendix B.